BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, DI KHAN

INTERMEDIATE (11TH CLASS) EXAMINATION, 2022

MATHEMATICS

Past Paper 2022 • Paper Code: DIK-HSSC-XI-2022 (A)

Retyped faithfully from the original board paper — same questions, no changes.

SECTION – A

Time Allowed: 20 min | **Maximum Marks:** 20 | **Paper Code:** DIK-HSSC-XI-2022 (A)

Q1. Choose the correct option i.e. A,B,C,D and fill it up in the correct bubble in the given MCQs sheet.

Qi. The period of -2 cos ecπx is_____

(A) 2
(B) 2π
(C) −2
(D) −2π

Qii. $[a_1 + (n - 1).r^{-1}$ is the rth term of _____ series.

(A) Arithmetic
(B) Geometric
(C) Arithmetico-geometric
(D) Arithmetic-harmonic

Qiii. The multiplicative inverse of z = 0 + 0i is _____

(A) 1 + 0i
(B) 0 + i
(C) 0 + 0i
(D) Doesn't exist

Qiv. If 1 + 2 + 3 + ...+ n = n(n+1)/2 then 1 + 2 + 3 ... ... + 24 =

(A) 1600
(B) 300
(C) 600
(D) 1300

Qv. $\vec{a} = i - j + k + k$ then $|\vec{a}|$ = _____

(A) [illegible in source scan]
(B) √2
(C) √3
(D) 3

Qvi. $\sin(\text{Tan}^{-1} x)$, $|x| < 1$ is:

(A) $x/\sqrt{(1-x^2)}$
(B) $1/\sqrt{(1-x^2)}$
(C) $1/\sqrt{(1+x^2)}$
(D) $x/\sqrt{(1+x^2)}$

Qvii. $\text{Tan}^{-1}[x/y] - \text{Tan}^{-1}[(x-y)/(x+y)] =$ _____

(A) π/2 (B) π/3

(C) π/4 (D) 3π/4

Qviii. $i^{57} + 1/i^{25} =$ _____

(A) i (B) 2i

(C) −i (D) 0

Qix. If z is a complex number then $z + \bar{z} =$ _____

(A) 2Im (z) (B) −2Re (z)

(C) 2Re(z) (D) 2Im (z)

Qx. $j \times (2j + 3k) =$ _____

(A) 3j (B) 2j

(C) 3i (D) 3k

Qxi. If $|\vec{a} + \vec{b}| = |\vec{a} - \vec{b}|$, then _____

(A) $\vec{a} \parallel \vec{b}$ (B) $\vec{a} \perp \vec{b}$

(C) $|\vec{a}| = |\vec{b}|$ (D) None

Qxii. If a,b,c form a H.P. then b = _____

(A) (a+c)/2 (B) 2/(a+c)

(C) 2ab/(a+c) (D) 2ac/(a+c)

Qxiii. If A = |1 6 7; 2 −5 8; 3 0 9| then $M_{13} =$ _____

(A) 13 (B) −15

(C) 26 (D) 15

Qxiv. If p(A) = 1/2 and p(B) = 0, then p(A / B) = _____

(A) 0 (B) 1/2

(C) 1 (D) Not defined

Qxv. The middle term in the expansion of $(2x + 5y)^4$ is _____

(A) $600\ x^2y^2$ (B) $120\ xy^2$

(C) $5000\ xy^3$ (D) $6x^2y^2$

Qxvi. $f : A \to B$ defined by $f(x) = x^2$ such that $A = \{-1,0,1,2\}$ and $B = \{0,1,4\}$ then f is _____

(A) One-one (B) Onto

(C) Bijective (D) Not a function

Qxvii. ${}^{n}P_{r} / {}^{n}C_{r}$ = _____ ?

(A) (n)!/(r)!
(B) (r)!/(n)!
(C) n!
(D) r!

Qxviii. The point of concurrency of the right bisectors of the sides of a triangle is called_____

(A) In-centre
(B) Circumcentre
(C) Orthocenter
(D) Centroid

Qxix. $\sin(\theta - \pi/2)$ = _____

(A) sin x
(B) −sin x
(C) cos x
(D) −cos x

Qxx. $f(x) = -2x^4 + \cos x$ is _____ function

(A) Odd
(B) Even
(C) Neither even nor odd
(D) Trigonometric

SECTION – B

Time Allowed: 2:40 Hrs | **Maximum Marks:** 50

Q2. Answer any TEN of the following parts. All questions carry equal marks.

i. Find the solution of $(z+1)^3 = -1$

ii. Show that $|bc\ ca\ ab;\ a\ b\ c;\ a^2\ b^2\ c^2| = |1\ 1\ 1;\ a^2\ b^2\ c^2;\ a^3\ b^3\ c^3|$

iii. The position vectors of the points A,B,C,D are $2i - j + k$, $3i + j$, $2i + 4j - 2k$, $-1 - 2j + k$ respectively. Show that AB is parallel to CD

iv. Find the sum of all multiples of 9 between 300 anti 700.

v. Draw the graph of $y = -4 + \sin x;\ 0 \le x \le \pi$

vi. If P(A) = 0.8, P(B) = 0.5 and p(B/A) = 0.4. Find $P(X \cup B)$

vii. By mathematical induction prove that $(5^{2n}-1)/24$ is an integer $\forall n \in N$

viii. Find the equation of quadratic function $y=ax^2+bx+c;\ a \neq 0$ which cuts the x-axis at points (−2,0) and (2,0) and also passes through the point (4,−6)

ix. If sin θ = [illegible in source scan] and terminal ray of θ lies in the second quadrant, then find cos θ/2

x. One diagonal of a parallelogram is 20cm long and at one end forms angles 20° and 40° with the sides of parallelogram. Find the length of the sides.

xi. For what value of the n is $(a^{n+1}+b^{n+1})/(a^n+b^n)$ [illegible in source scan]

xii. Verify that the triple product of i−j, j−k and k−i is zero.

xiii. Show that $\cos^{-1} x + \sin^{-1} x = \pi/2$

SECTION – C

Maximum Marks: 30

Note: Note: Attempt any Three Questions. All questions carry equal marks:

Q.3 Maximize F(x,y) = 2x + y subject to the constraints $x + y \leq 6$, $x + y \geq 1$, $x, y \geq 0$

Q.4 Solve by gauss elimination method. $x - y + 4z = -4$, $2x + 2y - z = 2$, $3x - 2y + 3z = -3$

Q.5 a. Find the coefficient of x^5 in the expansion of $(2x^2 - 3/x)^{10}$

b. Show that $\sin(\alpha+\beta).\sin(\alpha-\beta)=\cos^2\alpha$ [illegible in source scan]

Q.6 a. Prove that for any equilateral triangle $r:R:r_1 = 1:2:3$.

b. Find the value of $2.1\bar{5}$.